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A Conservative Phone-Level Pronunciation Coach for Standard Istanbul Turkish

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Abstract

Turkish pronunciation feedback remains underdeveloped compared with the English- and Mandarin-centred literature on automatic pronunciation assessment. This paper presents a conservative phone-level pronunciation coach for Standard Istanbul Turkish. The system combines a deterministic rule-based grapheme-to-phoneme (G2P) front end over a 49-phone inventory, a Turkish wav2vec 2.0 phone recogniser fine-tuned on a ∼275-hour training split of read speech, a multi-speaker phoneme atlas, and a runtime that aligns the target phone sequence against the recogniser's free decode. For every target phone it emits a status (`correct`, `incorrect`, or `missing`) and a continuous phonetic-distance score; targetless decoded phones are reported as `extra` rows, and acoustic typicality (`quality`) and duration (`length`) are reported as separate, non-gating channels. On 2,324 phones from two native Turkish speakers, the detection-first operating point catches 65.62% (42/64; Wilson 95% CI 53.4–76.1) of intended-error loci. This comes at flagged false-positive rates of 5.29% on control prompts and 5.54% on natural words, with 53.85% error-locus precision. A safety-first operating point that abstains on recogniser-unreliable contrasts reduces those false-positive rates to 0.22% and 0.80% and raises precision to 70.37%, but detection falls to 29.69%. The available phonetic-distance and recogniser-confidence summaries do not separate these false positives from genuine deviations. A same-protocol XLS-R-300M rerun yields the same intended-error detection (65.62%) with a similar false-positive profile. The central message is that recognition accuracy does not entail feedback safety; the two are traded against each other on the same recordings.

Keywords Turkish pronunciation assessment, grapheme-to-phoneme, wav2vec 2.0, phone recognition, mispronunciation detection, safety-detection frontier

1. Introduction

Pronunciation feedback is difficult to scale because a useful tutor must do more than transcribe speech. It must compare a learner's recording with a target pronunciation, decide whether a specific phone-level contrast is supported, and avoid giving confident but wrong feedback when the evidence is weak. Automatic pronunciation assessment and mispronunciation detection study this problem, but the public literature concentrates on a small set of learner languages (El Kheir et al., 2023; Zhang et al., 2021). For Turkish the problem is broader than a missing learner corpus. A phone-level coach needs three components that are all absent at once: a reliable

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target-IPA generator (G2P), an IPA-level phone recogniser, and a feedback layer that stays safe on natural speech. Turkish speech technology has mostly addressed automatic speech recognition for an agglutinative lexicon (Oflazer & Saraçlar, 2018; Sak et al., 2012), and diction work has relied on small perceptual studies rather than a reproducible phone-level system (Akdoğan, 2024; Gülen, 2018).

This paper presents a Turkish pronunciation coach for Standard Istanbul Turkish. The system converts the target text to an expected phone sequence, runs a Turkish phone recogniser on the recording, aligns the recogniser's free decode against the target, and measures acoustic evidence against a multi-speaker phoneme atlas. The per-phone feedback it returns keeps phone identity (the aligned status) separate from acoustic typicality (a separate `quality` channel). The central empirical finding is a *safety–detection frontier*: on natural correct speech the recogniser produces confident but spurious phone substitutions that are not separable from deliberate errors by the available score/confidence summaries, so lowering the false-positive rate necessarily lowers detection.

The contributions are:

1. A deterministic Turkish G2P front end over a 49-phone inventory plus a symbolic primary-stress mark, used for both phone-recognition labels and target templates, and validated against three independent benchmarks.
2. A phone-level coach runtime that aligns the target sequence to the recogniser's free decode and emits a target-phone status (`correct`, `incorrect`, `missing`) plus targetless `extra` rows with a continuous phonetic-distance score, while surfacing atlas-based acoustic typicality (`quality`) and a soft duration channel (`length`) as separate, non-gating evidence.
3. A two-speaker native validation protocol with three prompt groups — correctly pronounced controls (CTL), natural-word controls (W_NAT), and deliberate-error prompts (W_ERR) — covering 2,324 scored phones, with Wilson 95% CIs on the main aggregate rates.
4. A quantified safety–detection frontier with two honest operating points: detection-first catches 65.62% of intended-error loci while producing 5.29% false positives for CTL and 5.54% for W_NAT; safety-first reduces those rates to 0.22% and 0.80%, but detection falls to 29.69%.
5. A same-protocol XLS-R-300M robustness check showing that the feedback-safety profile is not materially changed by swapping the wav2vec backbone.

2. Related Work

Automatic pronunciation assessment has traditionally used phone-level posterior evidence, recogniser confidence, or learned scoring models to compare target and realised speech (Hu et al., 2015; Witt & Young, 2000). Recent work increasingly uses transformer and self-supervised speech representations, multi-task objectives, and acoustic-phonetic embeddings (Chen et al., 2024; Gong et al., 2022; Lin & Wang, 2021; Shahin et al., 2025; Ye et al., 2022). These systems motivate phone-level evidence, but they also highlight two practical issues for a tutoring product: evaluation metrics are not unified, and a recogniser's best transcription is not the same thing as safe corrective feedback.

wav2vec 2.0 and cross-lingual self-supervised models provide strong acoustic representations for low-resource phoneme recognition (Babu et al., 2022; Baevski et al., 2020). For Turkish, this is useful because large learner-pronunciation corpora are not available, while read-speech corpora can support a phone recogniser and a multi-speaker phoneme atlas (Ardila et al., 2020; Institute of Smart Systems and Artificial Intelligence (ISSAI), Nazarbayev University, 2022).

Turkish is a near-phonemic orthography (Göksel & Kerslake, 2005) with productive morphology, so a deterministic front end is a defensible engineering choice. Prior Turkish work supports sub-lexical modelling for the agglutinative lexicon (Oflazer & Saraçlar, 2018; Sak et al., 2012), and rule-based G2P is an established approach for Turkish computational pronunciation (Altinok, 2016). Turkish stress and prosody have their own descriptive tradition (Erdem Nas & Akbulut, 2021; Sezer, 1981), but the present system is scoped to segmental and local quantity feedback rather than full sentence-level prosody.

3. Method

3.a. System Overview

The runtime accepts a target utterance and a recording. It produces an expected phone sequence with G2P, runs a Turkish wav2vec recogniser to obtain a free phone decode with per-phone confidence, derives phone spans from the posterior path, measures acoustic features where the span is usable, and aligns the target sequence against the decoded sequence to emit a phone-level assessment.

Each target phone receives one status and a continuous score, plus two auxiliary, non-gating channels (Table 1).

Field	Question	Main evidence
`status`	Does the decoded phone match the target position?	Edit alignment of target phones against the free decode.
`score`	How far is the realised phone from the target?	Articulatory-feature phonetic distance.
`quality`	Is the realised phone acoustically typical where features are available?	Atlas Gaussian-mixture typicality; user-facing interpretation is currently trusted mainly for vowels.
`length`	Is the duration compatible with target vowel length?	Phone duration relative to local speech rate.

Table 1: Runtime output fields and evidence channels.

The runtime deliberately keeps these signals separate. A phone can be aligned as `correct` but acoustically atypical, or be `incorrect` while the duration channel still supports the target length. Quality and length never override the aligned identity; they are reported alongside it so the interface and the researcher can read them independently.

3.b. G2P Front End

The front end maps normalised Turkish text to a 49-phone inventory plus a symbolic primary-stress mark. The inventory includes eight short vowels, eight long vowels, the open /æ/ allophone, core consonants, context-sensitive allophones such as palatal stops and dark/light lateral realisations, and pedagogical allophones such as aspirated word-initial stops and the word-final devoiced tap.

The pipeline applies text normalisation, lexical exceptions, morphological analysis (a finite-state Turkish morphological analyser), grapheme-to-phone mapping, and a fixed sequence of 15 ordered phonological rules. The main processes include vowel harmony, ğ-conditioned lengthening or glide realisation, /e/ opening in closed sonorant contexts, palatal stop realisation before front vowels, nasal place assimilation, lateral allophony, and Sezer-style stress assignment (Sezer, 1981) for the relevant exception classes. The same front end creates phone labels for recogniser training and expected templates for assessment, which keeps training, evaluation, and runtime targets aligned. Algorithm 1 summarises the front end.

Algorithm 1. Rule-based G2P front end.

```
Input:  word w          Output: IPA phone sequence P
w ← NFC-normalise(w); trim edge punctuation
if w in abbreviation reference:  return parse(reference IPA)      # e.g.
TBMM, TRT; no rules
if w in exception lexicon:  P ← entry phones;  etype ← entry.type;  M ←
∅
else:
    P ← (w is acronym) ? letter-by-letter IPA : context-free
grapheme→IPA
    etype ← ∅;  M ← morphological analysis(w)            # root,
suffixes, POS, proper?
# fixed-order phonological rules (order preserves inter-rule
dependencies)
P ← vowel_harmony(P)
P ← soft_g(P, w, proper?)
P ← palatalisation(P)
P ← lateral_allophony(P, w, etype)
P ← v_allophony(P)
P ← nasal_place_assimilation(P)
P ← e_opening(P, w, etype)
if pedagogical_allophones:  P ← aspiration(P);  P ← tap_devoicing(P)
P ← stress(P, w, etype, M)
P ← canonicalise_stress(P)            # move stress mark to syllable
onset
return P
```

The frozen G2P engine is validated against three independent benchmarks (Table 2): an expert-curated 75-word set (Blind-75), a forced-alignment-derived dictionary of 41,373 Turkish entries (MFA v3) (McAuliffe et al., 2017), and a noisier Wiktionary scrape of 14,015 entries. The Blind-75 evaluation is blind in the operational sense: the rule engine was frozen before the test set was annotated by an independent expert, and 6 of the 13 symbol differences were attributable to gold-annotation errors and corrected across 5 rows. Seven segmental differences remain (across five categories), all accepted limitations of the rule-based design: lexical long vowels in Arabic/Persian loans, word-initial cluster vowel epenthesis, tap devoicing inside clusters, loan palatalisation without a lexical entry, and open-vowel over-application on a productive suffix.

Benchmark	Entries	PER	Exact match
Blind-75 (expert)	75	0.015	90.7%
MFA v3 dictionary	41,373	0.032	81.7%
Wiktionary IPA	14,015	0.118	39.1%

Table 2: G2P validation across three independent benchmarks.

3.c. Phone Recogniser

The recogniser is a wav2vec 2.0 model fine-tuned as a phoneme CTC head (Graves et al., 2006) over the 54-token vocabulary (49 phones, the symbolic primary-stress mark, and four CTC specials: `<pad>`, `<unk>`, the inter-word space `<spc>`, and `<blank>`). Two checkpoints are maintained: MMS-1B (default) (Pratap et al., 2024) and XLS-R-300M (alternative) (Babu et al., 2022). Both are fine-tuned on a ∼275-hour training split drawn from Mozilla Common Voice 24.0 and ISSAI TSC (Ardila et al., 2020; Institute of Smart Systems and Artificial Intelligence (ISSAI), Nazarbayev University, 2022). The pool totals roughly 310 hours after a 0.5–15 s duration filter, partitioned with a strictly speaker-aware 90/5/5 split and sentence-level deduplication between train and dev/test.

The CNN feature encoder is frozen during fine-tuning; only the transformer stack and a freshly initialised linear CTC head are updated. Both runs use an effective batch of 32, a learning rate of 1e-4, FP16 with gradient checkpointing, and seed 42 on a single RTX 3090; the best checkpoint is selected by minimum dev-set PER. The final checkpoint arguments record run-specific scheduling values: XLS-R-300M used per-device batch 4 with gradient accumulation 8, 500 warmup steps, and 400-step evaluation/checkpoint cadence; MMS-1B used per-device batch 8 with gradient accumulation 4, 1,000 warmup steps, and 500-step evaluation/checkpoint cadence. XLS-R-300M trained for 59,888 steps (8 epochs, ≈20 GPU-hours) and MMS-1B for 62,500 steps (7.4 epochs, ≈37 GPU-hours).

On the held-out test split (15,277 utterances, 661,008 reference phones) MMS-1B reaches 4.13% Full-IPA PER (3.62% segmental) and XLS-R-300M reaches 4.05% (3.49% segmental); the two are within 0.1 points. PER is the phone edit distance normalised by the reference phone count; Full-IPA PER scores all 49 phones plus the symbolic primary-stress mark, while segmental PER excludes that mark. Table 3 breaks the test-split PER down by backbone and source corpus.

Backbone	Full-IPA PER	Segmental PER	Common Voice	ISSAI TSC
MMS-1B (default)	4.13%	3.62%	4.00%	4.41%
XLS-R-300M	4.05%	3.49%	3.88%	4.41%

Table 3: Held-out test-split phone error rate (PER). Full-IPA and segmental PER are over the whole split; the Common Voice and ISSAI TSC columns give Full-IPA PER per source corpus.

In the coach, the recogniser supplies a *free* phone decode: the most likely phone sequence over the recording, with a per-phone confidence. The coach aligns this decode to the G2P target rather than scoring a target-conditioned posterior, so the status reflects what the recogniser actually decoded at each position. This is a deliberate design choice that keeps the runtime auditable: the free decode answers "what did the model hear?" directly. The trade-off is that the system inherits the recogniser's confident confusions on natural speech, which we analyse below.

Long vowels illustrate why the recogniser is not treated as the sole authority. Many lexical long vowels in Turkish occur in Arabic/Persian loans and require exceptional lexical knowledge; formal spelling may mark them with a circumflex, which can distinguish both pronunciation and meaning (`kar` "snow" vs. `kâr` "profit"). The acoustic training transcripts contain this cue only sparsely: in the acoustic training manifest, Common Voice has circumflex marks in 992/86,865 rows (1.14%; 775/63,974 unique transcripts), while ISSAI-TSC has none in 182,760 rows. We therefore keep vowel duration as a separate evidence channel rather than expecting the CTC recogniser alone to separate long and short vowels.

3.d. Phoneme Atlas

The phoneme atlas is a multi-speaker acoustic reference built from the broader Turkish read-speech pool — roughly 545 hours (463,003 segments) combining Mozilla Common Voice, ISSAI TSC, and public-domain audiobooks — with forced-alignment timing and per-phone feature extraction. Forced alignment is used only here, offline; at runtime the coach derives phone spans directly from the wav2vec posterior path, with no external aligner. For each phone observation it stores the G2P target phone, the recogniser-supported observed phone, the local feature span, formant estimates, duration relative to local speech rate, and speaker/session identifiers. Formant estimates (F1/F2/F3) are extracted with a Burg analysis (Praat via Parselmouth) and kept only where measurement is stable (Boersma & Weenink, 2026; Jadoul et al., 2018). Vowel typicality is summarised as per-phone Gaussian mixture models with a log-density calibration artifact; a soft duration channel supplies the length evidence: phone duration is converted to a z-score against the local speech-rate-normalised median for its phone class, reported as a soft note that never gates the status. The validation prompts are recorded separately and held out from the atlas pool. Each Gaussian component scores the observed feature vector with a Mahalanobis distance, and a held-out log-density calibration maps the raw likelihood to a typicality in $[0, 1]$, where 1 is class-typical and low values are atypical. This calibrated `quality` is surfaced as feedback, not as an additional penalty on the alignment-based score.

The coach uses the atlas in a limited way: it computes per-phone acoustic typicality where the feature vector is available, but the user-facing interpretation is currently trusted mainly for vowels, where formant evidence is stable enough to support feedback. Consonant quality values remain diagnostic analysis evidence rather than an independent user-facing decision. Missing or unreliable acoustic evidence is not treated as an error and does not lower the aligned status; it is simply reported as absent.

3.e. Assessment Policy

The assessment policy aligns the G2P target sequence against the free-decoded phone sequence with a Needleman–Wunsch global edit alignment (Needleman & Wunsch, 1970) (unit substitution, insertion, and deletion costs). Each target phone then receives one of four statuses: `correct` (matches the decoded phone at that position), `incorrect` (aligns to a different decoded phone), `missing` (no decoded counterpart), or `extra` (a decoded phone with no target). For aggregate safety metrics, `incorrect` and `missing` are counted as *flagged*; `extra` is not a target-phone false positive but lowers the word-level score through an extra-penalty term.

Each status carries a continuous phonetic-distance score $s \in [0, 1]$. An exact match scores 1. For a substitution, the score is the larger of the class floor $s_{\min}$ and one minus the weighted sum of differing articulatory features, where f ranges over articulatory features (for vowels: height, backness, rounding, length; for consonants: place, manner, voicing, aspiration, secondary articulation), and w_f is the per-feature penalty (e.g. 0.20 for vowel backness, 0.24 for vowel length, 0.12–0.46 for consonant manner/place). Vowel height is the one graded feature: its penalty is 0.12 per height step, capped at 0.24. A cross-class substitution collapses to $s = 0.05$, and a floor $s_{\min}$ of 0.30 (vowels) or 0.12 (consonants) bounds same-class penalties. This makes a single front/back vowel slip, such as /i/ realised as /ɯ/, score far higher than a cross-class deviation. By construction, phones with `missing` or `extra` status take $s = 0$, so the mean phone score reflects deletions and insertions, not only substitutions. Algorithm 2 states the alignment and per-phone scoring together.

Algorithm 2. Target–observation alignment, status, and phonetic-distance score.

```
Input:  target phones E[1..n], free-decoded phones O[1..m]
Output: per-phone rows (status ∈ {correct, incorrect, missing, extra},
score s ∈ [0,1])

# Global edit alignment: minimise edit cost, break ties by maximising
matches
dp[i,0] ← (i,0) "missing";   dp[0,j] ← (j,0) "extra"
for i ← 1..n, j ← 1..m:
    diag ← (E[i]=O[j]) ? (cost, matches+1, "correct") : (cost+1,
matches, "incorrect")
    dp[i,j] ← argmin over { diag, "missing" (cost+1), "extra" (cost+1) }
```

```
by (cost ↑, matches ↓)
rows ← backtrace(dp)
# Continuous phonetic-distance score per row
for each row (t = expected, h = observed):
    if t = ∅ or h = ∅:           s ← 0             # missing / extra
    else if h = t:               s ← 1             # exact
    else if class(t) ≠ class(h): s ← 0.05          # cross-class
    else:  s ← max(s_min, 1 − Σ_f w_f · [t,h differ on feature f])
           # s_min = 0.30 (vowels) / 0.12 (consonants); w_f as defined
above
return rows with statuses and scores
```

A contrast-authority table limits which contrasts an acoustic-analysis candidate may overwrite in the raw wav2vec observation before alignment; it adjusts only the observed phone for a small pre-registered set of contrasts and never emits a user-facing label. The policy therefore produces an explainable per-phone row — target, decoded phone, status, score, and (where available) quality and length — rather than a single correct/incorrect label.

4. Evaluation

4.a. Validation Set

The validation recordings are organised into three groups (Table 4).

Group	Meaning	Metric role
CTL	Correct control prompts	Flagged false-positive rate.
W_NAT	Naturally produced target words	False-positive rate under lexical variation.
W_ERR	Deliberate-error prompts with pre-registered intended error loci	Intended-error detection and error-locus precision.

Table 4: Validation prompt groups and their metric roles.

Both validation speakers are L1 Turkish and reported anonymously as K1 and K2. Each speaker reads the same 75-prompt list (75 prompts $\times$ 2 speakers = 150 recordings), producing 227 CTL, 749 W_NAT, and 186 W_ERR scored phones per speaker — 2,324 phones in total across the three groups. The recordings are held out from the phoneme atlas and from the wav2vec training, dev, and test partitions. The set is deliberately small and native-only: both speakers are known to produce the targets correctly, so the controls give an honest floor on false positives rather than a learner-population estimate. A target phone is *flagged* when its status is `incorrect` or `missing`. The prompt inventory contains 36 W_ERR prompts, but the metric denominator is the set of pre-registered intended error loci, not the prompt count: some prompts contain two intended loci and some are held for coverage/reading context, yielding 32 scored loci per speaker and 64 in total. Speakers are instructed to read the prompt word with the specified contrast deliberately mispronounced. A W_ERR locus counts as detected when that intended phone position is flagged; error-locus precision is the share of flagged W_ERR target positions that fall on an intended locus.

4.b. Aggregate Results

Group	Phones	Correct	Incorrect	Missing	Extra	Main rate (Wilson 95% CI)
CTL	454	430	18	6	1	flagged FP 5.29% (3.58–7.75)
W_NAT	1,498	1,415	63	20	10	flagged FP 5.54% (4.49–6.82)
W_ERR	372	294	73	5	23	detection 65.62% (42/64; 53.4–76.1)

Table 5: Two-speaker (native) validation summary. Rates carry Wilson 95% CIs.

The validation summary (Table 5) reports the detection-first operating point. It catches 65.62% (42 of 64 intended-error loci; Wilson 95% CI 53.4–76.1) of deliberate errors. The cost is a flagged false-positive rate of 5.29% (24/454; 3.58–7.75) on control prompts and 5.54% (83/1498; 4.49–6.82) on natural words, with error-locus precision of 53.85% (42/78; 42.9–64.5). Because the set has only two speakers, the Wilson score intervals (Wilson, 1927) are wide and accompany the main aggregate rates. The mean per-phone score stays high on natural-correct groups (0.968 CTL, 0.969 W_NAT) and falls to 0.916 on W_ERR, so the continuous score separates the deliberate-error set even where the status label is conservative.

4.c. Recogniser Backbone Robustness

The main feedback evaluation uses MMS-1B because it is the default runtime checkpoint and the source of the frozen frontier rows. To check whether the result is merely an MMS-1B artifact, the same 150 validation recordings were rerun through the same status/score protocol with XLS-R-300M. Although XLS-R has a slightly lower held-out PER, the feedback metrics remain essentially the same: intended-error detection is identical (65.62%), the CTL and W_NAT false-positive rates stay near 5%, and precision rises only marginally (Table 6).

Backbone	Role	CTL FP	W_NAT FP	Detection	Precision@err
MMS-1B	Default	5.29% (24/454)	5.54% (83/1498)	65.62% (42/64)	53.85% (42/78)
XLS-R-300M	Robustness	5.07% (23/454)	5.81% (87/1498)	65.62% (42/64)	56.00% (42/75)

Table 6: Recogniser-backbone robustness under the same two-speaker validation protocol.

The close match (Table 6) supports the paper's narrower claim: the safety problem is not resolved by choosing the alternative wav2vec checkpoint. The safety–detection frontier below is therefore reported on the MMS-1B result of record, while XLS-R is kept as a robustness check rather than a second deployment policy.

4.d. *Safety–Detection Frontier*

The detection-first operating point flags every alignment deviation. Its 5%-level false-positive rate raises an obvious question: can a safety-first policy suppress false positives without losing detection? The evaluation shows it cannot, for a structural reason: the false positives are not recogniser-uncertainty noise. The recogniser decodes them with high confidence — the median free-decode confidence is ≈0.98 on the 81 substitution false positives (32 vowel, 49 consonant; Table 8) versus ≈0.99 on intended-error loci — and they are phonetically close substitutions, such as nasal-to-tap, /s/-to-beta-fricative, and aspirated-/p/-to-/b/ confusions, that overlap the deliberate errors in both score and confidence. Missing-phone false positives have no decoded phone. The only signal that separates the two is the *contrast class*: caught intended errors are weighted toward vowel contrasts (front/back, rounding), where the recogniser is reliable, while many false positives are confident consonant confusions or missing-phone artifacts.

Table 7 reinterprets the same frozen recordings under policies that progressively abstain (route to a soft, non-flagged outcome) on allophonic, then consonantal, then sparse-evidence contrasts. Allophonic substitutions are surface variants that resolve to the same underlying phoneme: long/short vowels, open-vowel variants, palatal/velar stops, nasal place variants, dark-lateral variants, approximant /v/ variants, aspiration, and devoiced taps. Treating them as non-errors lowers the flagged false-positive rate from 5.29%/5.54% to 4.41%/4.67%. Flagging only non-allophonic vowel contrasts lowers it to 1.54%/2.14%, and additionally softening `missing` brings it to 0.22%/0.80%, below 1% on both correct-speech groups. But each gain costs detection, which falls from 65.62% to roughly 30%, because abstaining on consonant contrasts also abstains on the genuine consonant errors there. Re-admitting only the most distant consonant substitutions (phonetic-distance score below 0.30) recovers a little detection — to 32.81% — at a small false-positive cost (last two rows of Table 7).

Policy	CTL FP	W_NAT FP	Detection	Precision@err
Detection-first (no softening)	5.29% (24/454)	5.54% (83/1498)	65.62% (42/64)	53.85% (42/78)
Soften allophones	4.41% (20/454)	4.67% (70/1498)	51.56% (33/64)	50.77% (33/65)
Vowel contrasts only	1.54% (7/454)	2.14% (32/1498)	29.69% (19/64)	59.38% (19/32)
Safety-first	0.22% (1/454)	0.80% (12/1498)	29.69% (19/64)	70.37% (19/27)
Vowels + far consonants	2.20% (10/454)	2.54% (38/1498)	32.81% (21/64)	50.00% (21/42)
Same + soften missing	0.88% (4/454)	1.20% (18/1498)	32.81% (21/64)	56.76% (21/37)

Table 7: Safety-detection frontier over the same 150 recordings.

Figure 1 plots the same frontier. The shaded points mark the two deployment modes this paper treats as honest: detection-first, which maximises intended-error coverage, and safety-first, which keeps flagged false positives below 1% by abstaining on recogniser-unreliable contrasts.

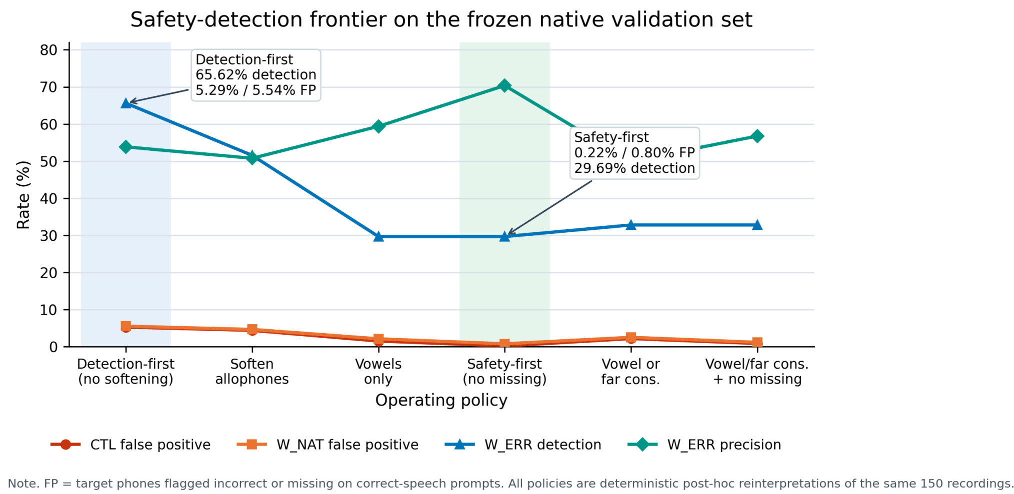

Figure 1: Safety–detection frontier over the frozen two-speaker native validation set. All policies are deterministic post-hoc reinterpretations of the same per-phone assessment rows.

This is the paper's central result: for natural Turkish speech, safe feedback is not obtainable by a threshold or a confidence gate; it is a trade against detection. Detection-first favours coverage; safety-first accepts the corresponding detection cost and restricts feedback to the contrasts where the recogniser is reliable.

4.e. Contrast Error Analysis

The contrast-level breakdown (Table 8) explains why the two modes diverge. In the correct-speech rows (CTL and W_NAT), 70.09% of false positives are consonant or missing-phone artifacts, while 61.90% of caught intended-error loci are vowel contrasts. Within the intended-error set, detection reaches 86.67% (26/30) on vowel loci but only 47.06% (16/34) on consonant loci, which is why abstaining on consonant contrasts gives up so much detection. The most repeated contrasts are compactly summarised here:

- Correct-speech false positives: nasal-to-tap (4), /b/-to-affricate (3), /o/ lengthening (3), and missing high-back unrounded vowel (3).
- Caught W_ERR loci: high-back unrounded vowel to [i] (6), front-rounded vowel to [o] (4), long /a/ to short /a/ (4), and /y/ to [u] (3).

These counts are computed from the per-phone validation rows.

Row group	Vowel	Consonant	Missing
Correct-speech false positives	32/107 (29.91%)	49/107 (45.79%)	26/107 (24.30%)
Intended W_ERR loci	30/64 (46.88%)	34/64 (53.12%)	0/64

Caught W_ERR loci	26/42 (61.90%)	16/42 (38.10%)	0/42
Detection within intended class	86.67% (26/30)	47.06% (16/34)	—

Table 8: Contrast-error class split over the frozen per-phone validation rows.

4.f. Speaker Breakdown

Speaker	CTL flagged FP	W_NAT flagged FP	W_ERR detection
K1	7.05% (4.4–11.1)	6.94% (5.3–9.0)	68.75% (22/32)
K2	3.52% (1.8–6.8)	4.14% (2.9–5.8)	62.50% (20/32)

Table 9: Speaker-level flagged false-positive and intended-error detection rates (Wilson 95% CI on the false-positive rates).

Even between two native speakers (Table 9) the safety profile varies with recording conditions. K1 shows a higher CTL flagged false-positive rate (7.05%) than K2 (3.52%), suggesting that the recogniser and atlas channels are sensitive to microphone and distance differences and misattribute them to phone identity. This between-speaker variation, observed even within L1 speech, supports the need for speaker adaptation and broader live-user data before any general-population claim.

4.g. Interpretation

The evaluation supports three claims. First, recognition accuracy (≈4.1% PER) does not translate into feedback safety: both MMS-1B and XLS-R-300M decode Turkish at low test PER, yet both produce roughly 5%-level false positives on natural-correct validation speech. Second, these false positives cannot be separated from genuine deviations by phonetic distance or recogniser confidence, so the safety–detection frontier is a structural limit, not a tuning oversight. Third, the status/score contract keeps this trade-off explicit and auditable, which is the prerequisite for choosing an operating point responsibly.

5. Discussion

The status/score design reflects a practical limitation of phone-level tutoring. Phone identity and acoustic typicality are related but not the same signal, and the recogniser's free decode is itself an imperfect witness on natural speech. Keeping the four channels separate — the aligned status, the phonetic-distance score, the atlas quality, and the duration channel — makes such mismatches visible to the reader instead of hiding them inside a single scalar. It is also what allows the safety–detection frontier to be measured at all.

The frontier reframes future work. Moving the safety-first point upward at a fixed false-positive budget is not a matter of threshold tuning. It requires a richer target-conditioned signal — for example a calibrated goodness-of-pronunciation margin or speaker-adapted acoustic typicality. Such a signal would need to separate confident recogniser confusions from real deviations, chiefly on the consonant contrasts where the free decode currently fails.

The current validation set is small and focused. It contains two native speakers and scripted prompt groups, not a large learner corpus with human teacher ratings. The results should therefore be read as system validation for Standard Istanbul Turkish feedback, not as a claim of general learner-population performance. Dialectal Turkish, child speech, impaired speech, noisy mobile capture, and full sentence-level prosody remain outside the validated envelope.

The system also deliberately does not claim a full prosodic tutor. Stress is carried only as a symbolic mark in the G2P target sequence and the CTC token set; the system does not produce an independent stress or prosody feedback channel. Discourse-level rhythm, pausing, intonation contours, and communicative fluency require a separate evaluation design.

5.a. Data, Code, and Reproducibility

The reference implementation — the deterministic G2P engine, the wav2vec fine-tuning and evaluation pipeline, the phoneme atlas tooling, and the runtime status/score engine — together with the validation prompt tables, compact evaluation summaries, configuration files, and figure-generation scripts is available as an open repository: https://github.com/onesvat/telaffuz-yz. The algorithms in this paper are derived from that implementation and are stated in enough detail to be re-implemented from the text alone. Large raw recordings, SQLite atlas databases, parquet manifests, and wav2vec checkpoints are treated as external artifacts under the repository's size policy; every headline number in this paper is reproducible from the committed small CSV/JSON/Markdown summaries.

6. Conclusion

This paper presented a conservative phone-level pronunciation coach for Standard Istanbul Turkish. The runtime combines a deterministic 49-phone G2P front end, a Turkish wav2vec 2.0 phone recogniser fine-tuned on read speech, calibrated acoustic reference statistics from a multi-speaker phoneme atlas, and an alignment-based assessment that emits target-phone statuses (`correct`, `incorrect`, `missing`), targetless `extra` rows, a continuous phonetic-distance score, and separate quality/length channels. On 2,324 phones from two native validation speakers, the detection-first operating point detects 65.62% of deliberate errors while producing flagged false-positive rates of 5.29% on CTL and 5.54% on W_NAT. The safety-first operating point drives those false-positive rates to 0.22% and 0.80% and raises precision to 70.37%, but detection falls to 29.69%. Repeating the same validation with XLS-R-300M leaves detection unchanged and produces the same roughly 5%-level false-positive profile, so the limitation is not removed by swapping the wav2vec backbone.

The immediate next step is broader validation: more human speakers, genuine learner recordings, teacher-rated feedback labels, and a target-conditioned signal able to move along the safety–detection frontier. The current contribution is the unified runtime, the evaluation frame, and the explicit quantification of why, for Turkish natural speech, recognition success is not the same thing as safe feedback. *References:*

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